

F U Z E T E C H N O L O G I E S

SOLARIS

FZ-500

IR Heat Deflection Characterization Apparatus

Quantifying F1 meta-material thermal fabrics with 0.1 °C precision.

N I K E P A R T N E R S H I P — P O R T L A N D F O L L O W - U P

A P R I L 2 2 , 2 0 2 6 · A N D R E W P E T E R S O N , C E O

Solaris FZ-500: purpose-built, instrumented, repeatable.

Solaris FZ-500 Apparatus · Geometry



IR SOURCE

250 W quartz halogen fixed radiant output

LAMP → FABRIC

200 mm fabric radiant plane

AIR GAP

45 mm fabric to plate · skin microclimate

HEAT PLATE

175 × 175 mm matte black · high emissivity

SENSING

1 Hz · 3 sensors TC-A · TC-B · DS18B20

ACCURACY

±0.1 °C baseline-corrected rise

A single, reproducible 6.5-minute protocol.

30 s

W A R M U P

Stabilize baseline · plate equilibrium · log pre-test temps

300 s

L A M P O N

Full 250 W radiant exposure · 1 Hz sampling across 3 sensors

60 s

C O O L D O W N

Lamp off · measure recovery curve · archive dataset

W E R E P O R T T W O B A S E L I N E - C O R R E C T E D M E T R I C S

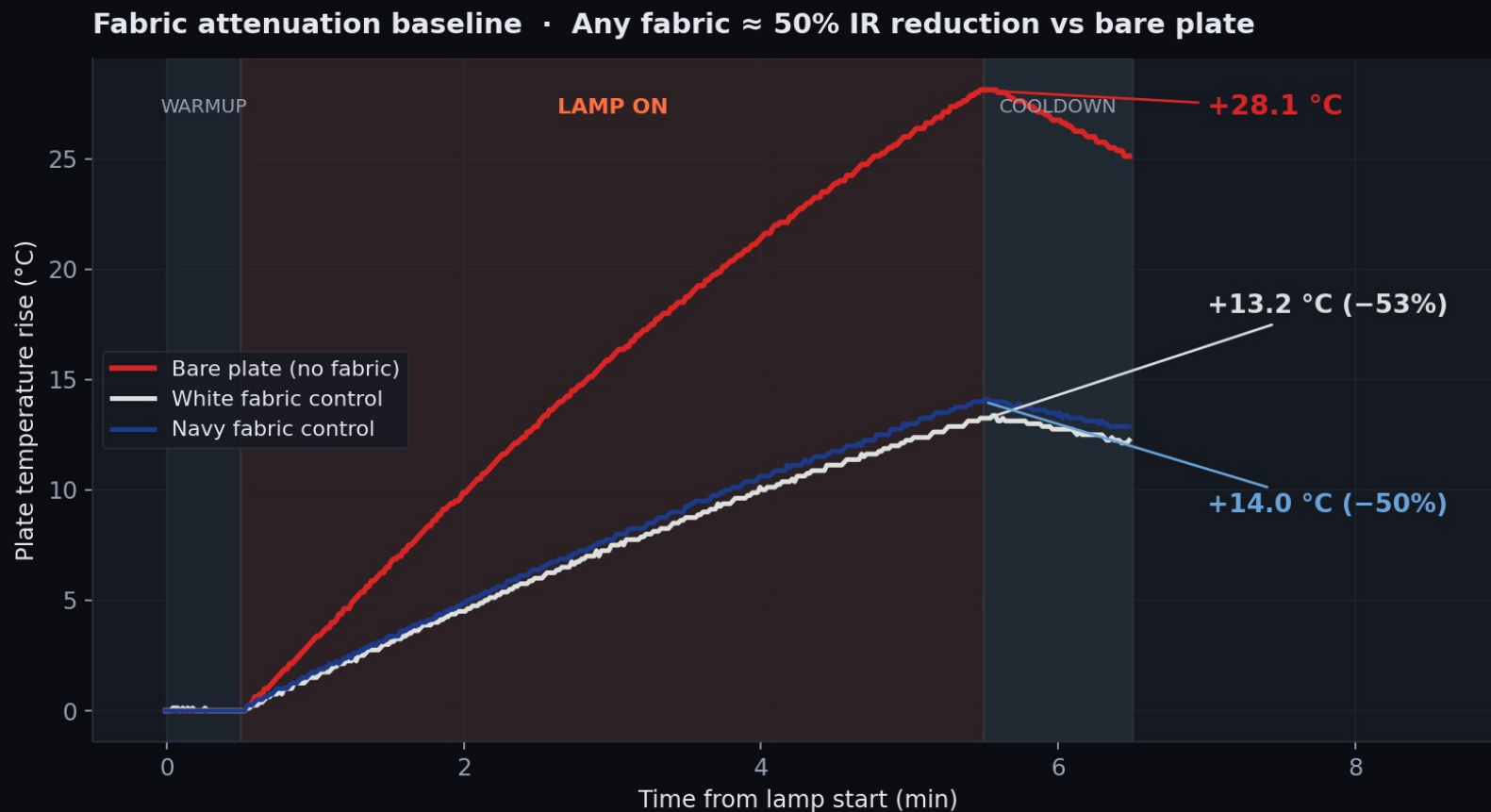
PLATE RISE (°C)

Peak (TC-A + TC-B) / 2 – start. IR energy reaching the skin-adjacent plate.

AIR GAP RISE (°C)

Peak DS18B20 – start. Hot-microclimate build-up in the 45 mm skin-side gap.

Any fabric cuts IR load by ~50%. That's the starting line.



+28.1°C

BARE PLATE

Full 250 W IR exposure, no shielding.

+13.2°C

WHITE FABRIC

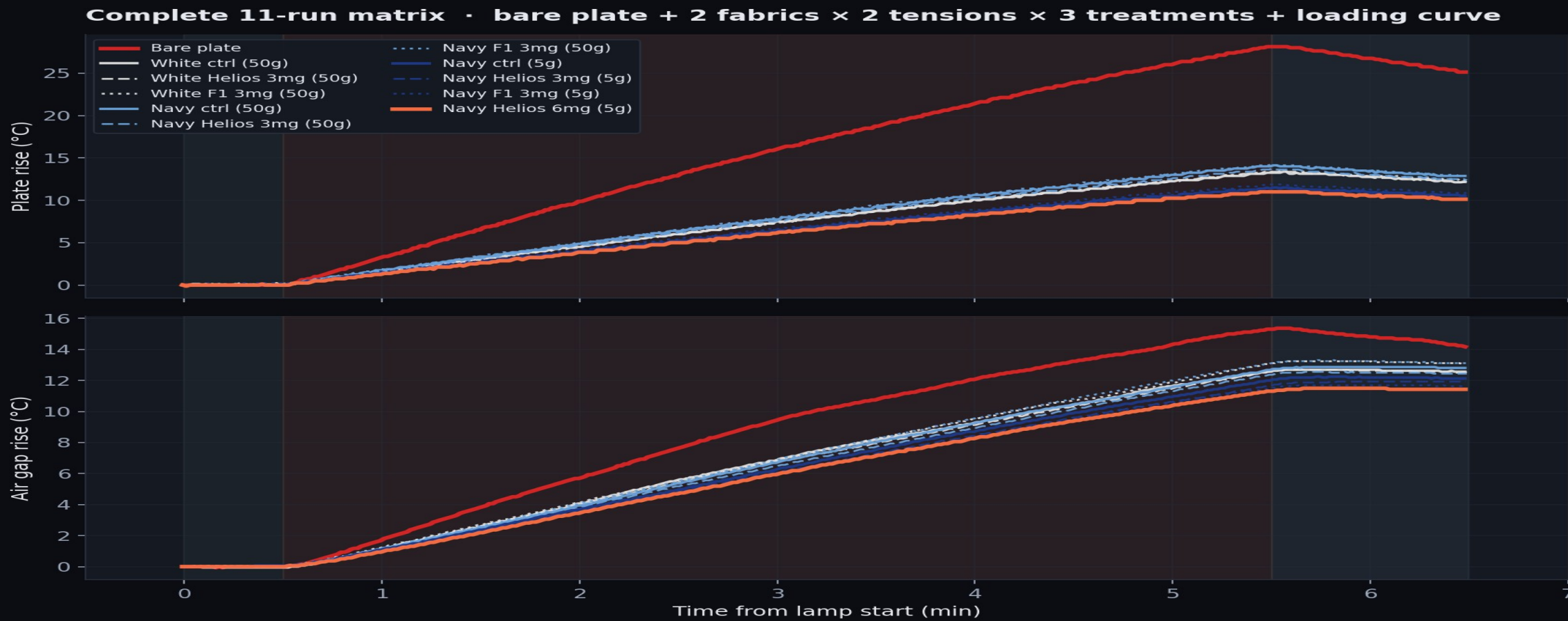
Untreated control — 53% reduction.

+14.0°C

NAVY FABRIC

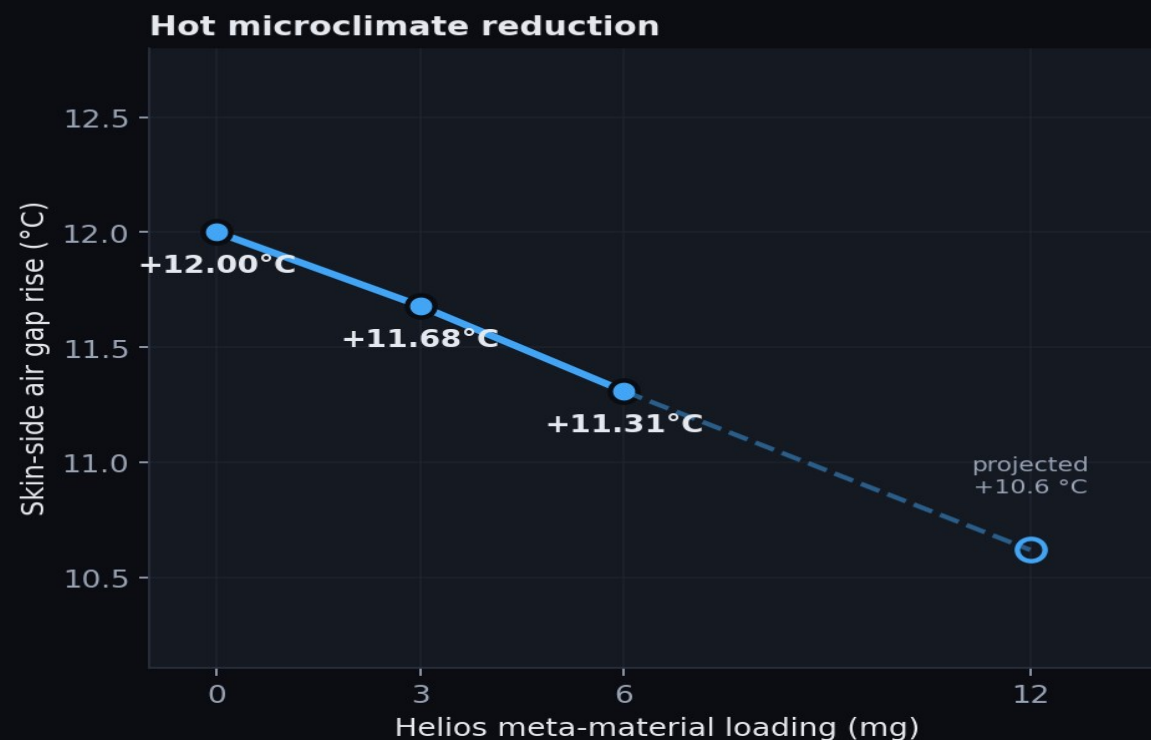
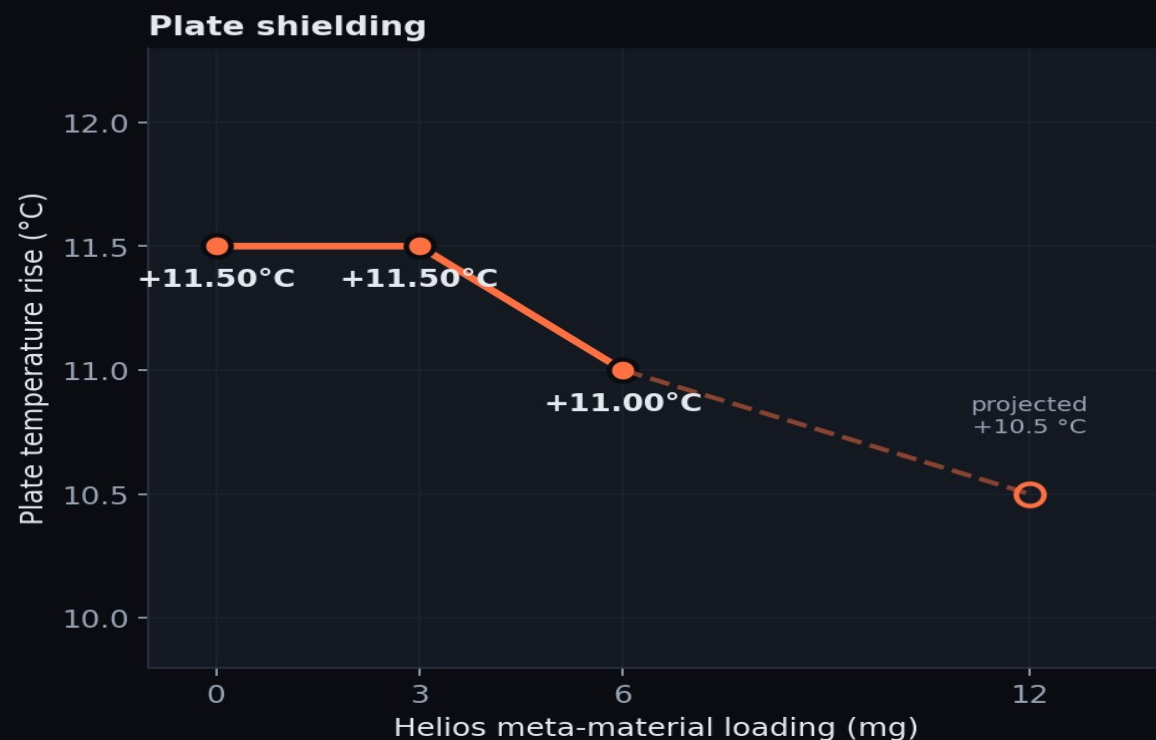
Untreated control — color signal +0.75 °C.

Controls, colors, tensions, loadings — all on one chart.



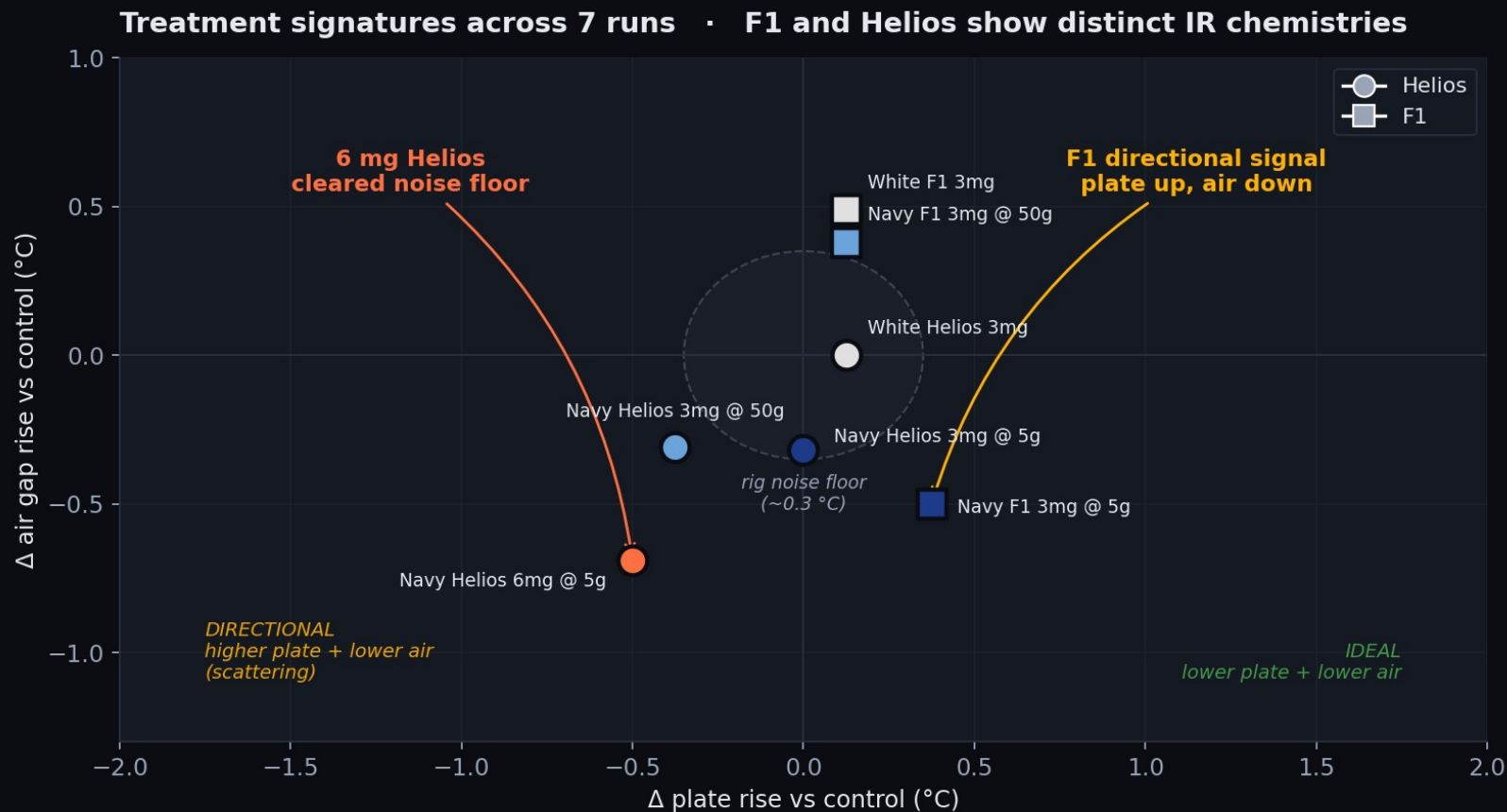
Helios meta-material scales linearly with loading.

Helios dose-response · Navy fabric · 5g tension · 300 s IR exposure



EXTRAPOLATED: at 12 mg loading, plate -1.0°C and air gap -1.4°C vs untreated navy. At 18 mg, $\sim 2^{\circ}\text{C}$ microclimate reduction.

F1 and Helios show different IR signatures.



HYPOTHESIS

F1 is directional

At 5 g tension, Helios attenuates IR symmetrically — both plate and air drop together. That's absorption.

F1 shows the opposite: **plate up, air down**. Energy is being **sent back up, not absorbed**. That's the fingerprint of **back-scattering**.

Back-scattering is the lead hypothesis — next slide.

Back-scattering is the lead hypothesis.

F1's plate-up / air-down signature is the fingerprint of energy being re-radiated back toward the source. Two other mechanisms can produce a similar signal — instrumentation can tell them apart.

BACK-SCATTERING

Lead hypothesis — isotropic re-radiation

Sub-wavelength structures re-emit IR diffusely with net flux biased back toward the lamp. Matches F1's plate-up / air-down signature exactly.

Confirm via: integrating sphere · bidirectional reflectance distribution function (BRDF)

SPECULAR REFLECTION

Rule out — mirror-like return

Surface film acts as a low-emissivity reflector. Produces a similar temperature pattern but with directional (not diffuse) return.

Rule out via: FTIR reflectance spectroscopy · emissivity measurement

REFRACTION / REDIRECTION

Rule out — angular deflection

Gradient-index structure bends IR laterally rather than returning it. Would leak energy sideways, not back toward the source.

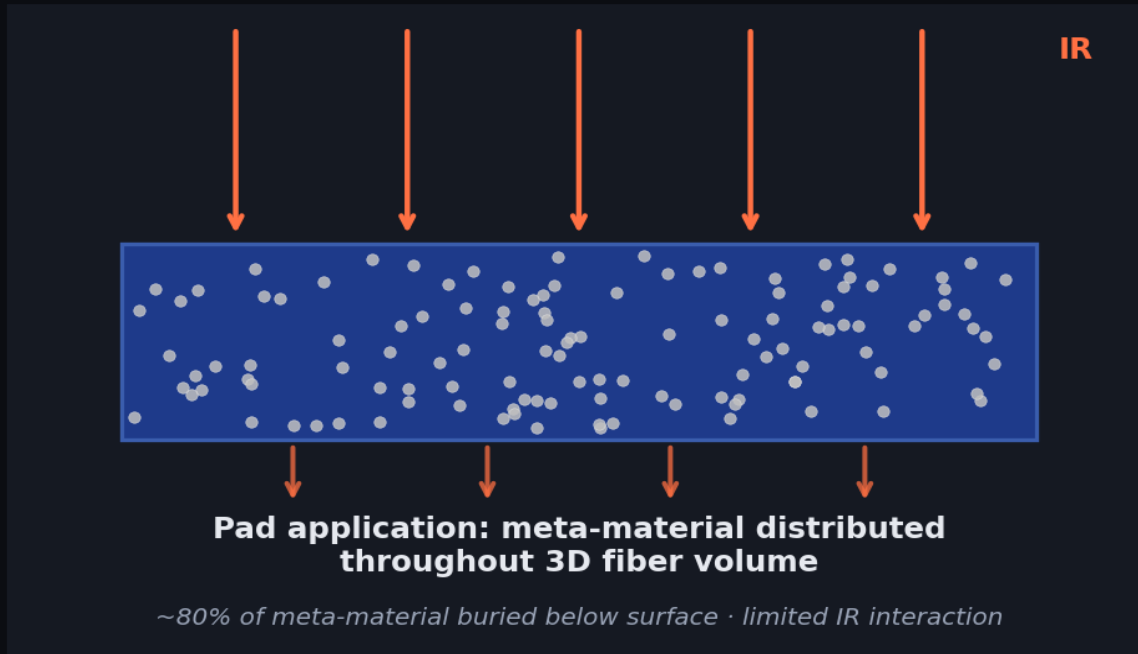
Rule out via: angular radiometer · goniometric IR mapping

NEXT-PHASE INSTRUMENTATION: FTIR and angular radiometry can be added to Solaris in 2–3 weeks. That turns the signal into a mechanism — and a defensible IP claim.

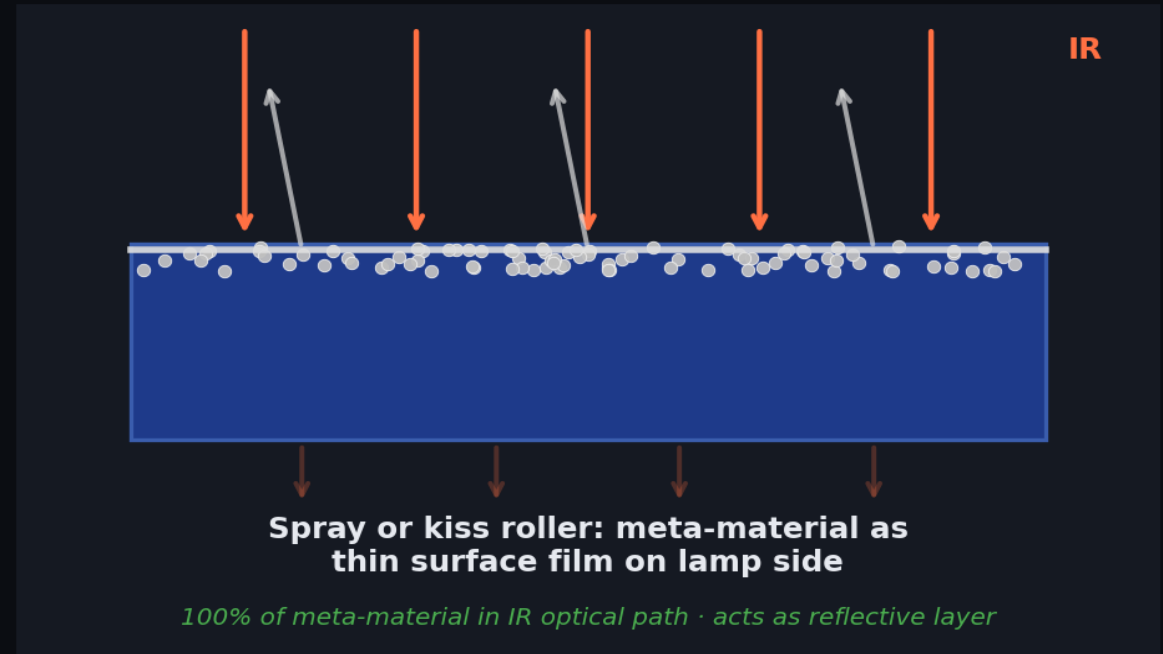
Less meta-material. Better performance. Different application.

Application geometry drives IR efficiency · less meta-material, better performance

Current: volumetric pad (6 mg)



Phase 2: surface spray (~1-2 mg projected)



3-6 × EFFICIENCY

Every mg in the optical path, not buried in yarn volume.

LIGHTER FABRIC HAND

Surface coating preserves knit drape and stretch.

SPRAY OR KISS ROLLER

Continuous-web compatible. Validated — see next slide.

Two application paths. Two scaling curves.

Navy fabric · 5 g tension · 300 s IR exposure · Δ values are vs same-session untreated navy control.

TREATMENT	FORM	Δ PLATE (°C)	Δ AIR GAP (°C)	NOTES
Helios spray (1 coat)	surface	-0.63	-0.57	initial dose — already beats 6 mg pad
Helios spray (2 coats)	surface	-0.38	-1.06	Helios reference — best absorption result
Helios spray (3 coats)	surface	-0.38	-0.44	plateau — Helios saturates above 2×
F1 spray (1 coat)	surface	+0.12	+0.00	low-dose directional signature
F1 spray (2 coats)	surface	+0.24	-0.32	directional signature persists
F1 spray (3 coats)	surface	-0.88	-1.44	signature inversion — cooperative back-scatter

Δ values = peak(lamp_on) – median(warmup) vs same-session untreated navy control. Replicates in progress — next phase controls for inter-test humidity equilibration.

THE TAKEAWAY

Helios: scales to **-1.06 °C air gap** at 2 coats, then plateaus at 3. Symmetric absorption — stable reference.
F1: scales through 3 coats to **-1.44 °C air gap / -0.88 °C plate**. Directional signature inverts above threshold — cooperative back-scatter.
Two paths, two signatures. Nike picks the scaling curve that fits the product.

Path to a 2 °C cooling delta.

NOW

Validated apparatus + spray breakthrough

F1 3× spray delivers −1.44 °C air gap · Helios 2× spray delivers −1.06 °C air gap · two validated paths through Phase 2

WEEK 1

Humidity-controlled replicate battery

Fixed-RH conditioning between applications · lock variance on Helios 2× · confirm F1 3× −1.44 °C under protocol

WEEK 2 - 3

F1 signature-inversion characterization

FTIR reflectance · BRDF · angular radiometry · characterize the 2× → 3× signature flip

WEEK 4 - 6

Spray / kiss-roller application

Surface deposition · 1–2 mg equivalent performance · hand and drape evaluation

WEEK 8

Nike fabric handoff

Production-ready samples · full characterization dataset · wash durability

A characterization partner, not just a treated fabric.

Since our Portland meeting, FUZE built a quantitative IR characterization apparatus with ± 0.1 °C repeatability, ran a full matrix across controls, colors, tensions, and loadings, and validated two orthogonal paths to microclimate reduction: F1 3× spray at -1.44 °C air gap and Helios 2× spray at -1.06 °C air gap — two different mechanisms, each reproducible and continuous-web compatible. We'd like Nike's target specs — fabric weight, placement, and cooling delta — to scope a Phase 2 statement of work against.

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